

REPLICA RESULT WITHOUT REPLICAS

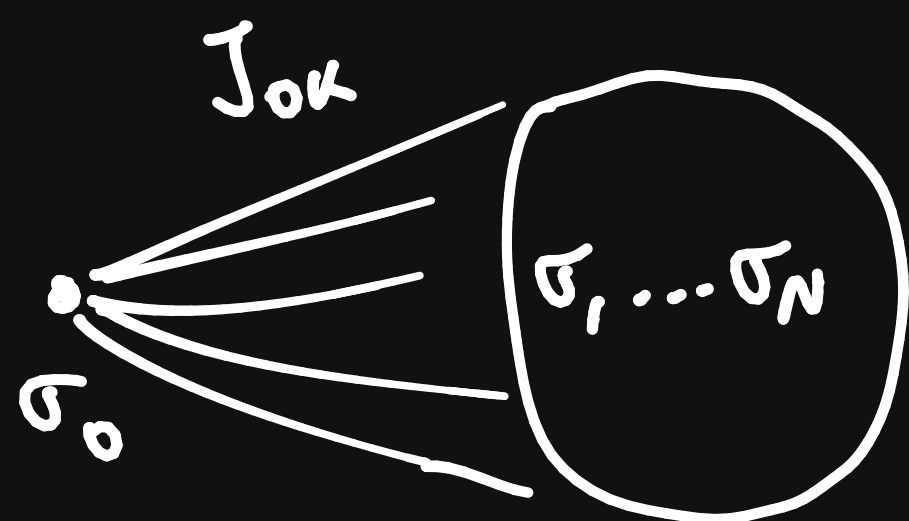
① CAVITY METHOD

①.1 THE FREE ENERGY SHIFT INSIDE A STATE

Consider N spins in equilibrium having the SK Hamiltonian (zero field)

$$H_N(\sigma) = -\frac{1}{\sqrt{N}} \sum_{i < j}^N J_{ij} \sigma_i \sigma_j \quad J_{ij} \sim \mathcal{N}(0, 1)$$

Next consider a new spin σ_0 and connect it to the N system spins with random couplings



$$J_{0k} \sim \mathcal{N}(0, 1)$$

The $N+1$ Hamiltonian is therefore $\overbrace{\sigma_0 h_0(\sigma)}^{\text{"CAVITY FIELD"}}$

$$H_{N+1}(\sigma, \sigma_0) = H_N(\sigma) - \overbrace{\frac{\sigma_0}{\sqrt{N}} \sum_k J_{0k} \sigma_k}$$

The partition function Z_{N+1} is therefore related to Z_N by

$$Z_{N+1} = \sum_{\sigma, \sigma_0} e^{-\beta H_{N+1}(\sigma, \sigma_0)} = \sum_{\sigma} e^{-\beta H_N(\sigma)} 2 \cosh(\beta h_0(\sigma))$$

$$= Z_N \langle 2 \cosh(\beta h_0(\sigma)) \rangle_N$$

where the average is done with respect to the Gibbs measure of the system with N spins.

Now imagine the Gibbs measure is split into several pure states

$$Z_N = \sum_{\alpha} Z_{\alpha(N)} \quad \text{with} \quad Z_{\alpha(N)} = e^{-\beta F_{\alpha(N)}}$$

Now condition the Gibbs measure in the previous result to a given state α

$$Z_{\alpha(N+1)} = \sum_{\sigma \in \alpha(N)} e^{-\beta H_N(\sigma)} 2 \cosh(\beta h_0(\sigma)) = Z_{\alpha(N)} \langle 2 \cosh(\beta h_0(\sigma)) \rangle_{\alpha(N)}$$

Similarly, the free energy shift in going from N to $N+1$ is

$$\Delta F_{\alpha} = F_{\alpha(N+1)} - F_{\alpha(N)} = -\frac{1}{\beta} \ln \langle 2 \cosh(\beta h_0(\sigma)) \rangle_{\alpha(N)} \quad \leftarrow \text{this is an intensive quantity !!}$$

(1.2) CAVITY FIELD INSIDE A STATE

Now consider the cavity field

$$h_o(\sigma) \equiv \frac{1}{\sqrt{N}} \sum_k J_{ok} \sigma_k$$

It depends on J_{ok} and on $\{\sigma_k\}$ that are in equilibrium between each other and those are independent quantities. We can compute the first two moments with respect to $\langle \rangle_{\alpha(N)}$ which are the only ones needed due to the CLT - the mean is

$$h_{\alpha(N)} = \langle h_o(\sigma) \rangle_{\alpha(N)} = \frac{1}{\sqrt{N}} \sum_k J_{ok} m_k^{\alpha(N)} \quad m_k^{\alpha(N)} \equiv \langle \sigma_k \rangle_{\alpha(N)}$$

Using the clustering property

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \sum_{i,j}^N \langle (s_i - m_i^{\alpha(N)}) (s_j - m_j^{\alpha(N)}) \rangle_{\alpha(N)}^2 = 0$$

which is valid as we are inside a state $\alpha(N)$ we have

$$\langle (h_o(\sigma) - h_{\alpha(N)})^2 \rangle_{\alpha(N)} = \frac{1}{N} \sum_{k,l} J_{ok} J_{ol} \langle (\sigma_k - m_k^{\alpha(N)}) (\sigma_l - m_l^{\alpha(N)}) \rangle_{\alpha(N)}$$

$$= \frac{1}{N} \sum_k \underbrace{J_{ok}^2}_{\sim 1} \left[\langle \sigma_k^2 \rangle_{\alpha(N)} - (m_k^{\alpha(N)})^2 \right] = 1 - q_1 \quad q_1 \equiv \frac{1}{N} \sum_k (m_k^{\alpha(N)})^2$$

Note that q_1 (overlap within a state) is not fluctuating anymore as it is for $h_{\alpha(N)}$ due to the J_{ok} . So prior to adding the spin σ the field distribution on state α on a newly added spin is Gaussian:

$$P_\alpha(h_o) = \frac{e^{-\frac{(h_o - h_{\alpha(N)})^2}{2(1-q_1)}}}{\sqrt{2\pi(1-q_1)}} \equiv \frac{1}{Z_{\alpha(N)}} \sum_{\sigma \in \alpha} e^{-\beta H_N(\sigma)} \delta(h_o - h_o(\sigma))$$

Therefore the free energy shift in state α is

$$\Delta F_\alpha = -\frac{1}{\beta} \ln \langle 2 \cosh(\beta h_o(\sigma)) \rangle_{\alpha(N)}$$

$$= -\frac{1}{\beta} \ln \int D h \ 2 \cosh(\beta h_{\alpha(N)} + \beta \sqrt{1-q_1} h)$$

$$= -\frac{1}{\beta} \ln \left[2 \cosh(\beta h_{\alpha(N)}) e^{\frac{\beta^2}{2}(1-q_1)} \right]$$

$$= -\frac{\beta}{2}(1-q_1) - \frac{1}{\beta} \ln(2 \cosh(\beta h_{\alpha(N)}))$$

$$= \langle \delta(h_o - h_o(\sigma)) \rangle_{\alpha(N)}$$

note that σ_o is not present here!!

"cavity field" field distribution on σ_o in absence of σ_o !

Notice the difference between the "empty field" distribution (Gaussian) and the "local field" distribution for the $N+1$ spin system at equilibrium:

$$P(h, \sigma_0) \propto \sum_{\sigma \in \pm 1} e^{-\beta H_N(\sigma) - \beta h_0(\sigma) \sigma_0} \delta(h - h_0(\sigma))$$

each N -spin configurations in state σ gets reweighted by σ_0 !!

$$\propto \left[\sum_{\sigma \in \pm 1} e^{-\beta H_N(\sigma)} \delta(h - h_0(\sigma)) \right] e^{\beta h \sigma_0} \propto e^{-\frac{(h - h_A(N))^2}{2(1-q_1)} + \beta h \sigma_0}$$

which is not Gaussian, as h is correlated to σ_0 (because of "Onnager" reaction σ_0 influences σ which in turn modify the field felt by σ_0). Indeed summing over σ_0 one finds

$$P_A(h) \propto e^{-\frac{(h - h_A(N))^2}{2(1-q_1)}} 2 \cosh(\beta h)$$

(1.3) SINGLE STATE ASSUMPTION - GETTING RS RESULTS

Now suppose there is only one state - We call the overlap within this state as

$$q_1 = q - W$$

We will denote by $h_A(N)$ simply by $\langle h \rangle_N$

This variable fluctuates only because of J_{0k} couplings:

$$\langle h \rangle_N \equiv \frac{1}{\sqrt{N}} \sum_k J_{0k} m_k^{(N)}$$

Due to the central limit theorem its distribution is Gaussian:

$$\overline{\langle h \rangle_N} = 0$$

$$\overline{\langle h \rangle_N^2} = \frac{1}{N} \sum_k (m_k^{(N)})^2 = q$$

The overlap of the state should satisfy the following self-consistent eq:

$$m_k = \int dh \sum_{\sigma_0} P(h, \sigma_0) \sigma_0 = \tanh(\beta \langle h \rangle_N)$$

$$N \equiv \int dh e^{-\frac{(h - h_A(N))^2}{2(1-q_1)}} 2 \cosh(\beta h) = 2 \cosh(\beta h_A(N)) e^{\frac{\beta^2}{2}(1-q_1)} \sqrt{2\pi}$$

$$\Rightarrow m_k(h) = \frac{1}{N} e^{-\frac{(h - \langle h \rangle_N)^2}{2(1-q)}} 2 \sinh(\beta h)$$

Then

$$q = \frac{1}{N} \sum_i \overline{m_k^2} = \int D h \tanh^2(\beta \sqrt{q} h)$$

which is the RS saddle point equation for the overlap.

The RS free energy can be written from the energy shift by noticing that in going from $N \rightarrow N+1$ we have also to add the J_{ij} variance from $\sqrt{N}$ to $\sqrt{N+1}$. One therefore finds that this can be taken into account by a rescaling of the temperature

$$\begin{aligned} \Delta F &= F_{N+1} - F_N = -\frac{1}{\beta} \ln Z_{N+1}(\beta \sqrt{\frac{N+1}{N}}) + \frac{1}{\beta} \ln Z_N(\beta) \\ &= -\frac{1}{\beta} \ln Z_{N+1}(\beta) + \frac{1}{\beta} \ln Z_N(\beta) - \frac{1}{\beta} \frac{1}{Z_{N+1}(\beta)} \frac{\partial Z_{N+1}}{\partial \beta} \frac{\beta}{Z_N} \end{aligned}$$

Next let's average over couplings:

$$\begin{aligned} \overline{\Delta F} &= -\frac{\beta}{2}(1-q) - \frac{1}{\beta} \int D h \ln 2 \cosh(\beta \sqrt{q} h) \\ &= (N+1)f - Nf + \frac{1}{2} \frac{\partial(\beta f)}{\partial \beta} \\ &= f + \frac{1}{2} \frac{\partial(\beta f)}{\partial \beta} = \frac{3}{2}f + \frac{\beta}{2} \frac{\partial f}{\partial \beta} = \frac{\partial}{\partial \beta} (\beta^3 f) \frac{1}{2\beta^2} \end{aligned}$$

This is an ODE:

$$\begin{cases} \frac{\partial}{\partial \beta} (\beta^3 f) = -\beta^3(1-q) - 2\beta \int D h \ln 2 \cosh(\beta \sqrt{q} h) \\ \beta f(\beta=0) = -\ln 2 \end{cases}$$

Then

$$\begin{aligned} \beta^3 f(\beta) &= -(1-q) \int_0^\beta dt t^3 - 2 \int_0^\beta dt t \int D h \ln 2 \cosh(t \sqrt{q} h) \\ \Rightarrow f(\beta) &= -\frac{1-q}{4} \beta - \frac{2}{\beta^3} \int_0^\beta dt \frac{dt^2}{dt} \int D h \ln 2 \cosh(t \sqrt{q} h) \\ &= -\frac{1-q}{4} \beta - \frac{1}{\beta} \int D h \ln 2 \cosh(\beta \sqrt{q} h) + \frac{1}{\beta^3} \int_0^\beta dt t^2 \int D h \tanh(t \sqrt{q} h) \sqrt{q} h \\ &= -\frac{1-q}{4} \beta - \frac{1}{\beta} \int D h \ln 2 \cosh(\beta \sqrt{q} h) + \frac{q}{\beta^3} \underbrace{\int_0^\beta dt t^3}_{\frac{\beta^4}{4}} (1-q) \end{aligned}$$

$$= -\frac{1-q}{4} \rho (1-q) - \frac{1}{\rho} \int D h \ln 2 \cosh(\rho \sqrt{q} h)$$

$$= -\frac{\rho}{4} (1-q)^2 - \frac{1}{\rho} \int D h \ln 2 \cosh(\rho \sqrt{q} h) = f_{RS}(\rho) \quad !$$

①.4 TAP EQUATIONS FOR A SINGLE STATE